MSc in Artificial Intelligence
Natural Language Processing Module

ASSIGNMENT 2

# Low-Resource Speech-to-Text Translation:

## Building an Irish→English Cascaded Pipeline

Module Weighting: 50% of total module mark
Submission Deadline: 10th May 2026, 23:59
Format: ACL-style conference paper + code submission
Dataset: IWSLT 2026 Irish–English Speech Translation Data
Mode: Individual submission

Students are expected to work independently. All submissions will be checked for academic integrity.

# 1. Overview

This assignment forms the second and major piece of assessed coursework for the Natural Language Processing module. It is worth 50% of your overall module grade. You will design, implement, and evaluate a cascaded speech-to-text translation system for the Irish–English language pair, using publicly available pretrained models and the IWSLT 2026 shared task dataset.

You are not required to train large neural models from scratch. Instead, you are expected to make informed choices among available pretrained components, integrate them into a coherent pipeline, rigorously evaluate the results, and write up your findings in the style of an academic conference paper using the ACL template.

# 2. Background and Motivation

## 2.1 Low-Resource Speech Translation

Speech translation (ST) — the task of converting spoken audio in one language into text in another — is challenging even for well-resourced language pairs. For low-resource languages, the challenge is compounded by limited transcribed audio data, fewer high-quality pretrained models, and greater linguistic diversity. Irish (Gaeilge) is one such language: a living minority language with active speaker communities but comparatively little digital data compared to languages such as French or Mandarin.

This assignment asks you to tackle a real, contemporary shared task: IWSLT 2026 Track 1 (Low-Resource Speech-to-Text Translation). Rather than competing in the full shared task, you will engage with the same data and evaluation framework but within the scope of an MSc research exercise.

## 2.2 Why Cascaded ASR→MT?

There are two broad architectural strategies for speech translation: end-to-end systems (which jointly model speech and translation) and cascaded systems (which chain a separate Automatic Speech Recognition (ASR) module with a Machine Translation (MT) module). Cascaded systems have several practical advantages in low-resource settings:

- ASR and MT components can be independently pretrained on large datasets in their respective domains.
- Errors can be diagnosed and improved at each stage independently.
- Strong off-the-shelf models (e.g. Whisper for ASR, NLLB for MT) are readily available.

This assignment focuses on the cascaded approach, requiring you to make principled decisions about each component.

# 3. Learning Outcomes

On successful completion of this assignment, you will be able to:

### Technical Skills

- Load and preprocess real-world speech data in a low-resource NLP setting.
- Deploy and integrate pretrained ASR and MT models using the Hugging Face ecosystem.
- Build a functioning end-to-end cascaded speech translation pipeline.
- Apply standard evaluation metrics (BLEU, chrF++) correctly to ST outputs.

### Experimental and Analytical Skills

- Design a principled comparison between a baseline and an improved system.
- Perform error analysis to identify and classify failure modes.
- Interpret evaluation results in the context of low-resource NLP challenges.

### Academic Communication

- Write a research paper following ACL conference-paper conventions.
- Communicate technical decisions and experimental findings clearly and critically.

# 4. Assignment Task

You must complete all of the following components:

## 4.1 Data Preparation

- Clone and explore the IWSLT 2026 Irish–English dataset from GitHub (https://github.com/acl-org/acl-style-files)
- Understand its structure: audio files, transcriptions, and reference translations.
- Apply any necessary audio preprocessing (resampling, format normalisation).
- Document dataset statistics: number of utterances, total duration, split sizes.

## 4.2 Baseline Cascaded Pipeline

- Select a pretrained ASR model suitable for Irish speech (e.g. a Whisper variant).
- Select a pretrained MT model capable of Irish→English translation (e.g. NLLB-200).
- Integrate the two components into a sequential pipeline.
- Run inference on the test or development set and save outputs.

## 4.3 Improved System

- Design and implement at least one meaningful improvement over the baseline.
- Possible directions include: a different ASR front-end, Irish-specific ASR fine-tuning, different MT model, beam search tuning, output post-processing, or language-model rescoring.
- Clearly document what you changed and why.

## 4.4 Evaluation

- Compute BLEU and chrF++ scores for both systems.
- Report results in a clearly formatted table.
- Optionally compute COMET scores if compute budget allows.

## 4.5 Error Analysis

- Analyse at least 20 system outputs qualitatively.
- Identify and categorise error types (e.g. ASR substitutions, deletion/insertion, MT mistranslations, hallucinations).
- Discuss which errors are most common and their likely causes.

# 5. Technical Requirements

### 5.1 Permitted Models and Tools

You are encouraged to use any of the following pretrained model families, but you are not limited to them:

- **Whisper (openai/whisper-*), Wav2Vec2, or Irish-specific ASR models from Hugging Face.**
- **NLLB-200 (facebook/nllb-200-*), Helsinki-NLP Opus-MT, or mBART-50.**
- **Hugging Face transformers, datasets, evaluate, jiwer, sacrebleu.**

You must document every external model used, including its Hugging Face model ID or source URL, licence, and the reason for choosing it.

### 5.2 Compute Constraints

All experiments can be runnable on Google Colab (free tier or Colab Pro) or a standard laptop with a GPU (or CPU with patience). You will not require access to multi-GPU clusters. Specifically:

- Do not train large models from scratch (e.g. no training of full Whisper or NLLB-200 weights).
- Fine-tuning lightweight adapters or LoRA layers on small subsets is permitted, but is not required.
- If your experiments required paid compute, state this clearly in your report and ensure your code can produce representative results within Colab free-tier constraints on a subset.

## 6. Deliverables

You must submit all of the following via the module submission portal:

| Item | Description | Format |
|---|---|---|
| Report | ACL-format paper describing your methodology, experiments, and findings | PDF (ACL template) |
| Code Notebook | Colab or Jupyter notebook containing all experiments | .ipynb |
| README | Short README explaining how to run your code and reproduce your results | .md or .txt |
| Results Files | BLEU/chrF++ scores and system outputs (hypotheses file) | .txt / .json |
| AI Use Declaration | Declaration of any AI tools used and how | Included in report appendix |

Your code must be reproducible. Include all random seeds and version information. A marker should be able to clone your notebook and reproduce your headline results.

## 7. Report Guidelines

### 7.1 Format

Your report must use the ACL 2025 LaTeX or Word template. The template is available at: https://github.com/acl-org/acl-style-files

Recommended length: 6–8 pages (excluding references and appendices). Appendices may include additional tables, error examples, and your AI use declaration.

### 7.2 Recommended Structure

- **Abstract (150–200 words): What you did, how, and your key result.**
- **Motivation, problem statement, your approach, paper structure.**
- **Briefly situate your work relative to prior approaches in ASR, MT, and cascaded ST.**
- **Dataset statistics, audio format, preprocessing steps.**
- **Baseline and improved pipeline architectures with justifications.**
- **Evaluation setup, metrics, and experimental conditions.**
- **Tables and discussion of BLEU/chrF++ scores.**
- **Qualitative analysis of system outputs.**
- **Interpretation of results, limitations, and potential improvements.**
- **Summary of contributions and findings.**
- **ACL-style bibliography.**

## 8. Evaluation Metrics

### Required Metrics

- **Standard translation quality metric. Use SacreBLEU for reproducibility.**
- **Character-level F-score metric; more robust than BLEU for morphologically rich languages.**

### Optional Metrics

- **Neural translation evaluation metric. Use unbabel-comet if compute allows.**
- **Word Error Rate on ASR transcriptions (if you have reference transcriptions).**

### Practical Diagnostics

- Coverage: proportion of input utterances that produce non-empty output.
- Repetition rate: frequency of degenerate repeated outputs.
- Length ratio: average ratio of hypothesis length to reference length.

## 9. Marking Scheme

The assignment is marked out of 100 and weighted at 50% of the module total.

| Category | Marks | Criteria |
|---|---|---|
| Technical Implementation | 30 | Baseline pipeline is correctly implemented and documented (10). Improved system demonstrates a genuine, justified difference (10). Code quality, reproducibility, and documentation (10). |
| Experimentation Quality | 20 | Appropriate and justified model selection (5). Experiments are clearly described and reproducible (5). Fair, well-structured comparison between systems (10). |
| Evaluation Rigour | 20 | BLEU and chrF++ computed correctly using SacreBLEU (10). Qualitative error analysis is substantive and insightful (5). Additional diagnostics used appropriately (5). |

| | | |
|---|---|---|
| Analysis and Discussion | 20 | Results interpreted critically, not merely reported (10). Limitations acknowledged honestly and specifically (5). Conclusions are well-supported by evidence (5). |
| Academic Writing | 10 | ACL format followed correctly (3). Writing is clear, structured, and professional (4). Figures, tables, and references correctly formatted (3). |
| Total | 100 | |

## 10. Distinction-Level Expectations

A distinction (typically 70%+) will typically demonstrate several of the following:

- Comparison of more than one ASR front-end (e.g. Whisper-small vs. Whisper-medium, or Whisper vs. a Wav2Vec2 Irish model).
- Comparison of more than one MT approach (e.g. NLLB-200 vs. Opus-MT or mBART-50).
- A well-motivated improvement with clear ablation or justification.
- A systematic, categorised error analysis with illustrative examples and discussion.
- Awareness of the limitations of BLEU/chrF++ in low-resource settings.
- A written discussion that situates results within the broader IWSLT context.
- Polished, publication-ready writing with correctly formatted citations.

## 11. Academic Integrity and AI Use

You are permitted to use AI writing and coding assistants (e.g. ChatGPT, Claude, Copilot, Gemini). However, the following rules apply:

- You must disclose any AI tool use in an appendix of your report, specifying what was used and how.
- All code and written content must be understood, verified, and edited by you. Do not submit unverified AI-generated code.
- All experimental results must be genuinely your own. Do not fabricate or adjust numbers.
- The intellectual analysis, interpretation, and critical discussion must be your own work.

Plagiarism detection tools will be applied. Undisclosed AI use, or submission of work that is substantially another's, constitutes academic misconduct and will be dealt with under university policy.

## 12. Submission Instructions

Submit all materials as a single ZIP archive via the module submission portal on Blackboard/Moodle. Name your archive as follows:

```
studentID_NLP_Assignment2.zip
```

The archive should contain:

- report.pdf — your ACL-format report

- notebook.ipynb — your Google Colab / Jupyter notebook
- README.md — brief setup and reproduction instructions
- results/ — directory containing hypothesis files and metric scores

Late submissions will be penalised at 5 marks per day up to a maximum of 5 days, after which a mark of zero will be awarded, in line with university late work policy. Extensions must be requested via the school office before the deadline.

Questions? Post to the module discussion board on Moodle. Please do not email for questions that are relevant to all students.